

AKSHAYA THOUTAM

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Languages/Tools: Python, R, Bash, JavaScript, HTML, CSS, Jupyter, NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch, TensorFlow, Azure, Linux/Unix, GitHub, Shiny

EDUCATION

HARVARD MEDICAL SCHOOL

Cambridge, MA 2025–2026

M.S in Clinical Research - 4.0 GPA

Capstone: “Transcriptional Programs Underlying Malignant Cell States in Cholangiocarcinoma: A Systematic Review and Meta-analytic Synthesis of scRNA-seq Evidence”

GEORGIA INSTITUTE OF TECHNOLOGY

Atlanta, GA 2020–2023

B.S in Neuroscience, minor in health and medical science

Advisor: Greg Gibson, Ph.D.

PUBLICATIONS

Co-first Author †

1. Alan DenAdel, Madeline Hughes, **Akshaya Thoutam**, Anay Gupta, Andrew W. Navia, Nicolo Fusi, Srivatsan Raghavan, Peter S. Winter, Ava P. Amini, and Lorin Crawford. Evaluating the role of pre-training dataset size and diversity on single-cell foundation model performance. *Nature Methods*. 23, 1447–1457 (2026).
2. Zeinab Navidi, **Akshaya Thoutam**, Madeline Hughes, Srivatsan Raghavan, Peter S. Winter, Lorin Crawford, and Ava P. Amini. Adaptive Resampling For Improved Cellular Representation Learning in Imbalanced Single-cell Samples. (2025). *Under peer review*.
3. Ajay Nadig & **Akshaya Thoutam**†, Madeline Hughes, Anay Gupta, Andrew W. Navia, Nicolo Fusi, Srivatsan Raghavan, Peter S. Winter, Ava P. Amini, and Lorin Crawford. Consequences of training data composition for deep generative models in single cell biology (2025). *Under peer review*.
4. Greg Gibson, John D. Rioux, Judy H Cho, Talin Haritunians, **Akshaya Thoutam**, Maria T Abreu, Steven R Brant, Subra Kugathasan, Jacob L McCauley, Mark Silverberg, Dermot McGovern. Eleven Grand Challenges for Inflammatory Bowel Disease Genetics and Genomics. *Inflammatory Bowel Diseases*. 2025 Jan 6;31(1):272–284.
5. Nasrin Hooshmand, **Akshaya Thoutam**†, Max Anikovskiy, Hagar I Labouta, Mostafa El-Sayed (2021). Localized Surface Plasmon Resonance as a Tool to Study Protein Corona Formation on Nanoparticles. *Journal of Physical Chemistry C*, 125(45), 24765–24776
6. **Akshaya Thoutam**, Narasaiah Kolliputi (2021). Epigenetics of pulmonary diseases. In T. Tollefsbol (Ed.), *Medical Epigenetics* (2nd ed., pp. 185–195). Elsevier. ISBN: 9780128239285
7. **Akshaya Thoutam**, Mason Breitzig, Richard Lockey, Narasiah Kolliputi (2020). Coronavirus: a shift in focus away from IFN response and towards other inflammatory targets. *Journal of Cell Communication and Signaling*. 2020.Dec;14(4):469–470

SOFTWARE

- 1. **scFM-dataselection:** Evaluating the role of pre-training dataset size and diversity on single-cell foundation model performance
- 2. **scFM-datamix:** Consequences of training data composition for deep generative models in single cell biology
- 3. **sc-AR:** Adaptive Resampling For Improved Cellular Representation Learning in Imbalanced Single-cell Samples

AWARDS/HONORS

BroadHacks 2025 Broadie’s Choice Award Winners Broad Institute	June 2025
College of Sciences Dean’s Scholar Georgia Institute of Technology	Awarded every semester
Early Research Award Georgia Institute of Technology	August 2021
Robert Gunn Student Award (Finalist) American Physiological Society	April 2023

LEADERSHIP

Models, Inferences, Algorithms (steering committee) Broad Institute	July 2025 – Present
EWSC Spatial transcriptomics study group (Organizer) Broad Institute	July 2026 – Present
Down to Dance (Founder/President) Georgia Institute of Technology	August 2020 – May 2023
Urban Health and Wellness committee leader GT AMSA	September 2020 – May 2021
Fundraising committee leader GT AMSA	September 2020 – May 2021

TALKS

- 1. Ex Vivo Joint Lab Meeting, **Microsoft Research** — State-Associated Drug Screening: Systematically Defining Dependencies Associated with Cell State Signatures. December 2025.
- 2. **GT Bioinformatics** and Quantitative Biosciences Panel — Career Mentoring: Industry Research and Research Internships. October 2023.
- 3. **Akshaya Thoutam** & Greg Gibson (2023, April). Pan-cancer atlas of tumor-infiltrating B cells. Oral presentation. Georgia Institute of Technology Undergraduate Research Symposium

RESEARCH EXPERIENCE

Project Ex Vivo BROAD INSTITUTE OF HARVARD/MIT	(July 2025 – Present)
Computational Associate II	
▪ Building novel batching strategies for single-cell foundation model training. ▪ Using multimodal machine learning with scRNA-seq and spatial transcriptomics data to define and engineer cell states in cancer.	

Biomedical Machine Learning – Project Ex Vivo MICROSOFT RESEARCH	(Feb 2024 – July 2025)
Predoctoral Research Assistant	
▪ Manipulating single cell foundation models to increase data depth and biology in capturing cell-state level biology ▪ Creating an end-to-end pipeline for alignment and analysis of single-cell RNA-seq data	

Levin Lab (Genomics) BROAD INSTITUTE OF HARVARD/MIT	(May 2023 – September 2023)
Computational Associate Intern	
▪ Integrated long-read MAS-seq data, Visium spatial transcriptomics data, and single-nucleus RNA-seq to investigate spatially organized RNA isoform diversity in a pilot study on Parkinson’s disease. ▪ Analyzed single-nucleus RNA-seq data of a SCN2A knockout mouse cortex from the SCHEMA Schizophrenia initiative.	

Knott Lab (Genomics + ML) | CEDARS-SINAI

Computational Associate Intern

(May 2022 – October 2022)

- Analyzed temporal and spatial differences of cell types with scRNA-seq and ATAC-seq data from patients of an ER+ Breast Cancer clinical trial to characterize the tumor microenvironment in three different time periods to elucidate the molecular mechanisms that drive ER+ breast cancer.

Gibson Lab | GEORGIA TECH

Computational Research Assistant

(May 2021 – December 2023)

- Curated 133,488 B cells characterized from 246 tumor samples across 9 human cancer types to identify distinct molecular characteristics in the tumor microenvironment across cancer types - creation of pan-cancer atlas of B cells.
- Compared the immune transcriptional spectrum across healthy colon, IBD, Crohn's Disease, and colon cancer progression.

Laser Dynamics Lab | GEORGIA TECH

Research Assistant

(May 2021 – May 2022)

- Computationally modeled and synthesized gold and silver plasmonic nanoparticles of different morphologies for their applications in biosensing and photothermal therapy in cancer treatment

Kolliputi Lab (Immunology) | UNIVERSITY OF SOUTH FLORIDA

Research Writer

(May 2020 – March 2021)

- Compared respiratory viruses MERS-CoV, SARS-CoV-1, HPIV3, RSV, IAV to SARS-CoV-2 to uncover monocyte cytokine production as a driving feature of COVID-19 infection.
- Invited to write the chapter on 'The Epigenetics of Pulmonary Disease' for the textbook Medical Epigenetics, 2E (Elsevier)

POSTERS Co-first Author †

1.

Akshaya Thoutam†, Madeline Hughes (2025, April). Evaluating the role of pre-training dataset size, diversity, and composition on single-cell foundation model performance. [RECOMB 2025 Research in Computational Molecular Biology](#).
2.

Akshaya Thoutam†, William Watson, Mohan William, Regina Nwosu (April 2023). 5HT2c against lorcaserin on spontaneous nerve root 3 activity in crayfish model. [Georgia Tech Undergraduate Neuroscience Symposium](#)
3.

Hira Anis, Akshaya Thoutam, Makeda Hailu, Vishal Dhere, Greg Gibson, Erin Connolly (2023, April) Single-cell Pan-cancer atlas reveals diversity and plasticity of tumor-infiltrating plasma cells. [Atlanta Workshop for Single Cell Omics 2023](#)
4.

Akshaya Thoutam† C Greg Gibson (2022, October). A Pan-cancer Single-cell Transcriptional Atlas of Tumor-infiltrating B cells. [National Symposium for Undergraduate Research](#)
5.

Akshaya Thoutam†, Erin Connolly, Greg Gibson (2022, April). A Pan-cancer Single-cell Transcriptional Atlas of Tumor-infiltrating Plasma Cells. [Atlanta Workshop for Single Cell Omics 2022](#).
6.

Akshaya Thoutam†, Erin Connolly, Greg Gibson (2022, April). A Pan-cancer Single-cell Transcriptional Atlas of Tumor-infiltrating Plasma Cells. [Atlanta Workshop for Single Cell Omics 2022](#).

SERVICE

Peer Review ICLR Gen ² workshop	Jan 2026 – March 2026
Discovery center tour guide Broad Institute 50 hours	July 2025 – Present
Teaching Harvard Medical School 50 hours	July 2025 – May 2026
Emergency Department Massachusetts General Hospital 74 hours	August 2024 – December 2024
Arts in medicine Program Emory Winship Cancer Center 150 hours	August 2020 – May 2021
Care package Advent Health Florida 100 hours	March 2020 – December 2020
Recycling initiative Cedars Sinai Hospital 200 hours	May 2022 – August 2022

SHADOWING

Pediatric Pathology | Boston Children's Hospital | **60 hours**

Pediatric Heme Oncology | Boston Children's Hospital | **8 hours**

Neurosurgery | Atlanta VA Medical center | **80 hours**

Pediatrician | All Pediatrics Tampa | **100 hours**

Sarcoma Oncology | Moffitt Cancer Center | **40 hours**

Thoracic surgery | Moffitt Cancer Center | **40 hours**